



Universität
Zürich^{UZH}

Departments of Geography and Chemistry

DSA101.1

Introduction to Data Analytics and Python

Prof. Alexandria Deliz Liang

Prof. Meredith C. Schuman, meredithchristine.schuman@uzh.ch

Who am I?

Meredith C. Schuman (Merry)

Born in Minnesota, USA

2012 PhD in Chemical Ecology, FSU,
Jena, DE

2012-2019 Group Leader, Max Planck
Institute for Chemical Ecology, Jena, DE

2019-2026 Assistant Professor
Since 2026 Associate Professor for
Spatial Genetics of Ecosystems, UZH, CH



**University of
Zurich^{UZH}**

**Departments of
Geography and Chemistry**

**We want to know why living beings vary and where their
variation matters for ecology and for people.**



Who are you?

Motivation



A Boeing Delta II 7425 expendable launch vehicle lifts off with NASA's Mars Climate Orbiter on Dec. 11, 1998

<https://www.simscale.com/blog/nasa-mars-climate-orbiter-metric/>



Motivation

The retracted paper analyzed excitatory synapse number and neurotransmitter release in mouse hippocampal neurons missing a presynaptic cell-adhesion molecule called neuexin-2. Südhof, professor of molecular and cellular physiology and neurosurgery at Stanford University, received the Nobel Prize for some of his previous work on neuexin proteins.

The authors retracted the study “because re-analysis of the original raw data for Figs. 2, 4 and 6 revealed that, although our analyses of the original data are supportive of the conclusions of the paper, unresolvable differences exist between these raw data and the published data source file that cannot be corrected by a simple erratum,” according to the 5 March [retraction notice](#).

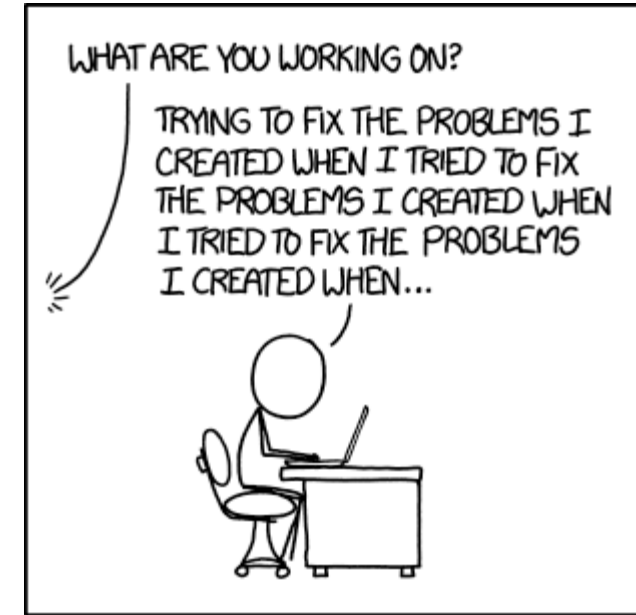
We want to solve a problem using a computer.

There will be many similar, but not identical problems.

Did we solve the original problem correctly?

What if...

- The data change?
- The assumptions change?
- We have 10x as much data? (Or 0.1x as much data?)
- We pass this on to a colleague, or to ourselves six months or six years from now?



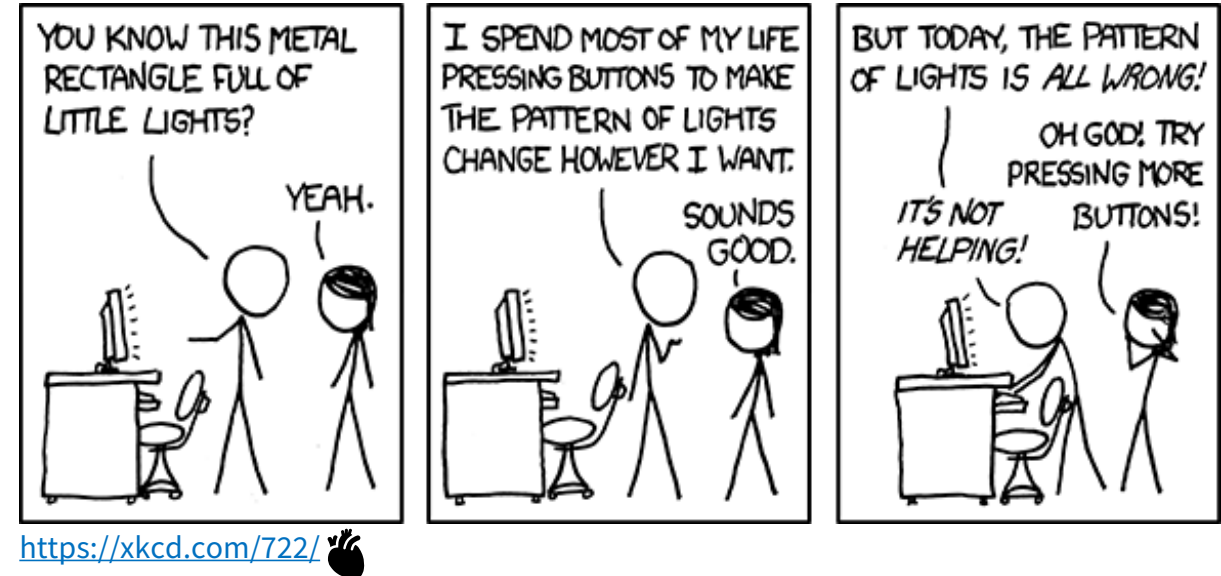
<https://www.xkcd.com/1739/>



We want to solve a problem using a computer.

If we made a mistake, what happens?

- Program crashes?
- Everything is *increeeedibly* slow?
- It still works for *this specific case*?
- The output is a *little bit* wrong?
- The output is wrong *sometimes*?

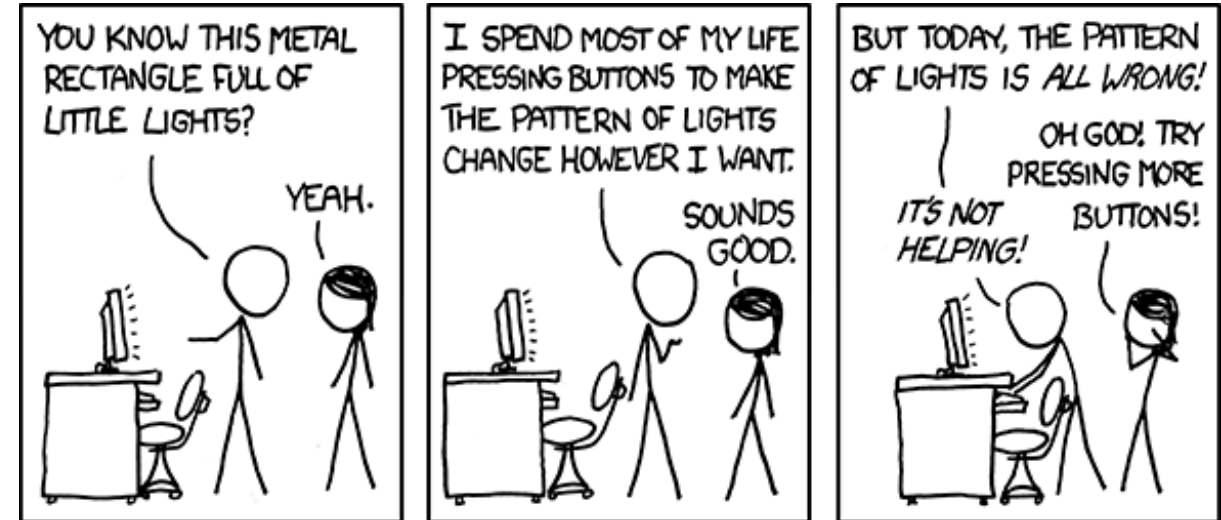
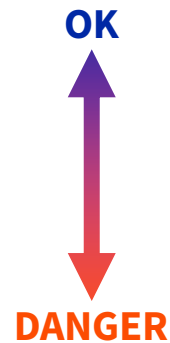


We want to solve a problem using a computer.

If we made a mistake, what happens?

Clear data science ranking:

- Program crashes?
- Everything is *increeeedibly* slow?
- It still works for *this specific case*?
- The output is a *little bit* wrong?
- The output is wrong *sometimes*?



<https://xkcd.com/722/> 🍷

LIKELIHOOD YOU WILL GET CODE WORKING
BASED ON HOW YOU'RE SUPPOSED TO INSTALL IT:



What is Git?

Git is a program for version control.

What is version control and why is it important?

- You change a file → change can be tracked
- A collaborator changes a file → changes can be merged
- You can go back and forth in time of your project
- You can branch out, test something and either integrate or revert changes
- **You can hardly break anything**

<https://docs.gitlab.com/ee/topics/git/>



The program
git logo by Jason Long, [CC BY 3.0](#)

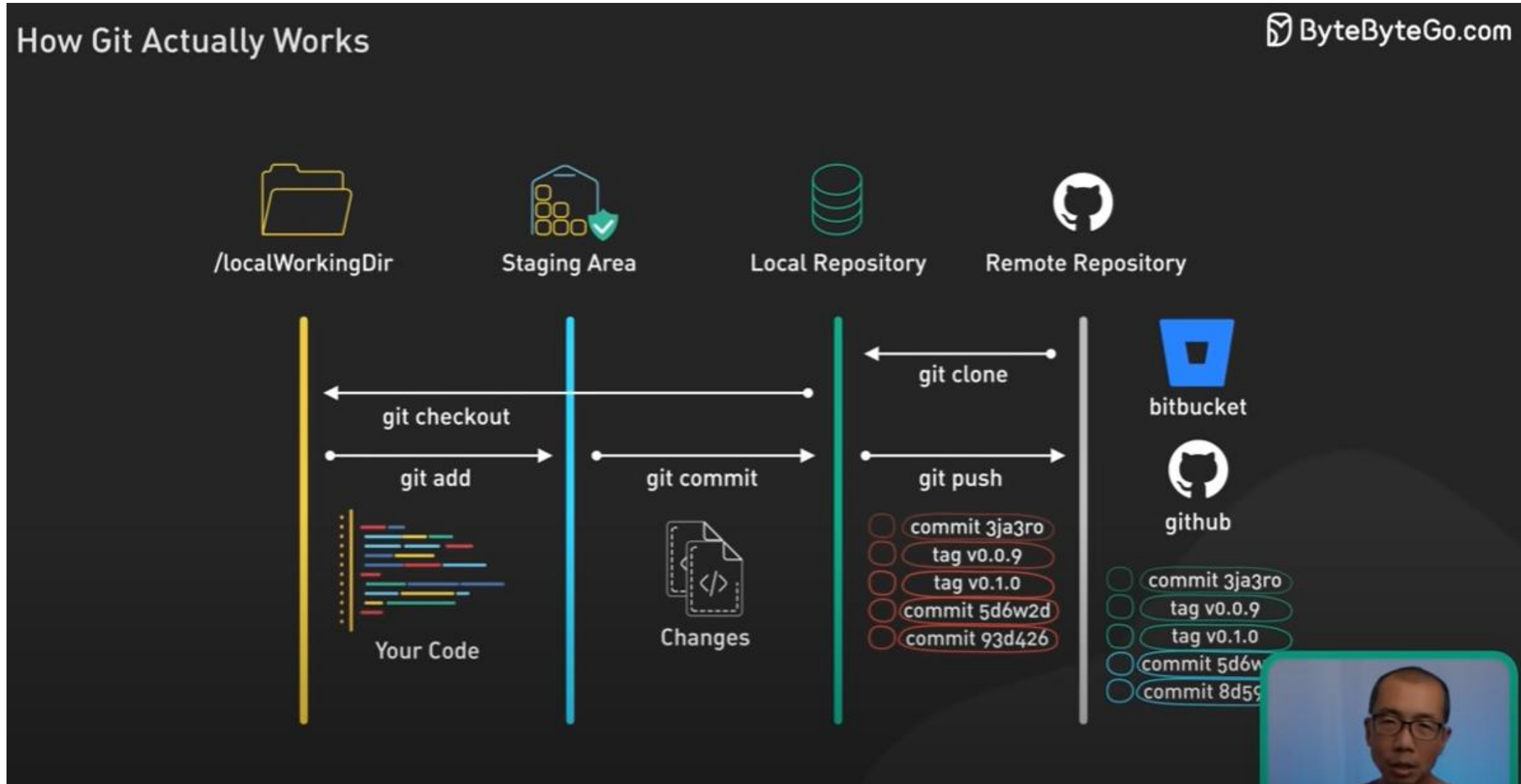


Institutional accounts,
European interface



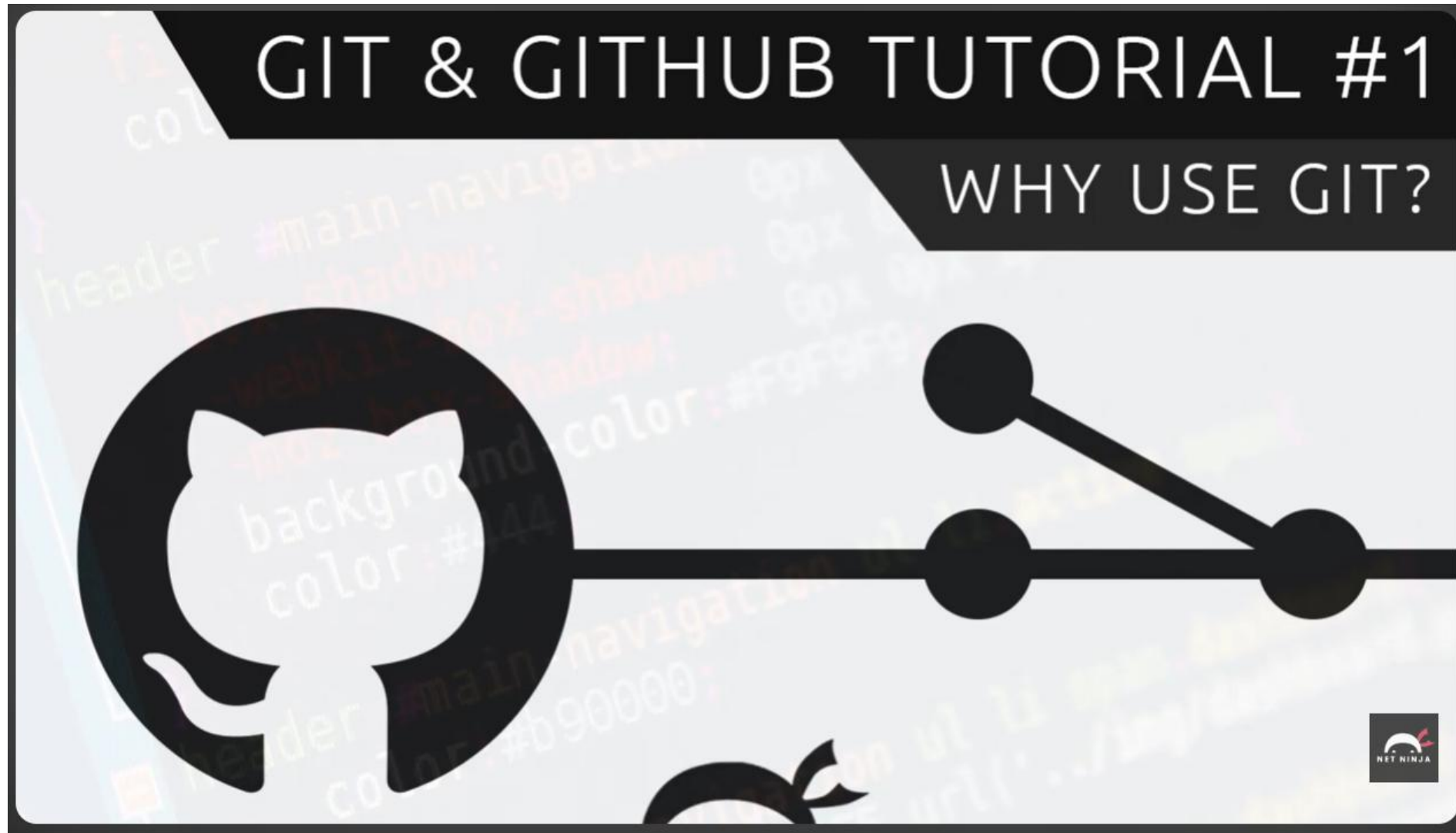
Free individual accounts,
North American interface
used worldwide

How git and GitHub work



https://youtu.be/e9lnsKot_SQ?si=g0d_pQD2eUPpP8Ae

How to git and GitHub



Please watch these GitHub [tutorials](#) before we next meet!

You may also find it helpful to refer to the [Git Cheat Sheet](#)

We will be working from a GitHub repo

Platform Solutions Resources Open Source Enterprise Pricing

Search or jump to... / Sign in Sign up

mcschuman / DSA101 Public

Notifications Fork 17 Star 1

<> Code Issues Pull requests 1 Actions Projects Security Insights

main 1 Branch 0 Tags

Go to file

<> Code

 mcschuman Updated main ReadMe file

f65016e · 35 minutes ago 25 Commits

collider	Re-organized repository to allow different coures narratives	38 minutes ago
exercise1	Add files via upload	last week
exercise2	Re-organized repository to allow different coures narratives	38 minutes ago
exercise3	Re-organized repository to allow different coures narratives	38 minutes ago
solutions	Re-organized repository to allow different coures narratives	38 minutes ago
.gitignore	Create .gitignore	last week
LICENSE	Initial commit	last week
README.md	Updated main ReadMe file	35 minutes ago

README AGPL-3.0 license

DSA101

Repository for DSA101 exercises (UZH DSA minor)

There are some exercises to get started and to introduce specific concepts, and a simulation to try out collaborative

About

Repository for DSA101 exercises (UZH DSA minor)

Readme

AGPL-3.0 license

Activity

1 star

0 watching

17 forks

Report repository

Releases

No releases published

Packages

No packages published

Languages

Jupyter Notebook 97.9%

Python 2.1%

Let's start!

- (1) [Make a GitHub account](#)
if you don't have one

github.com/signup

Research communic...Think Check Attend -NotebookLMAdobe Acrobat

All Bookmarks

Create your free account

Explore GitHub's core features for individuals and organizations.

See what's included



Already have an account? [Sign in](#)

Sign up for GitHub

Continue with Google

Continue with Apple

or

Email*

Password*

Username*

Your Country/Region*

Email preferences

Create account

Password should be at least 15 characters OR at least 8 characters including a number and a lowercase letter.

Username may only contain alphanumeric characters or single hyphens, and cannot begin or end with a hyphen.

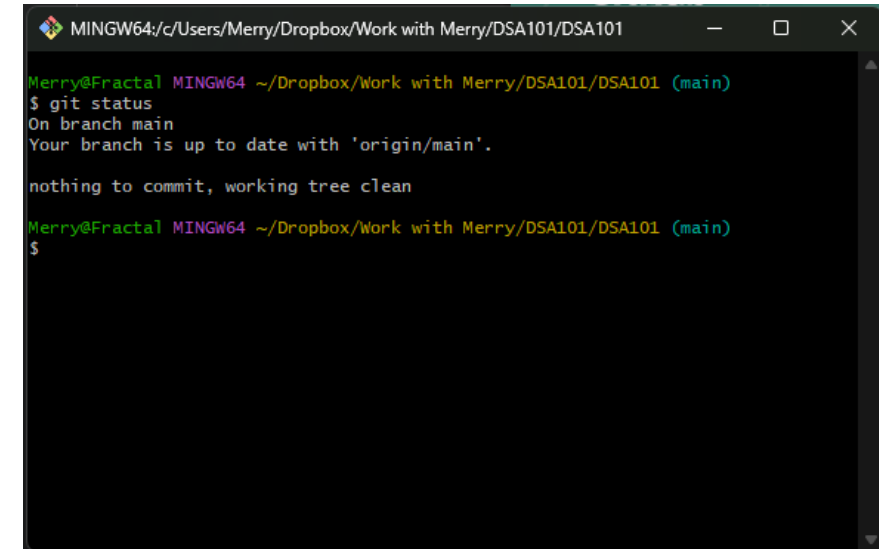
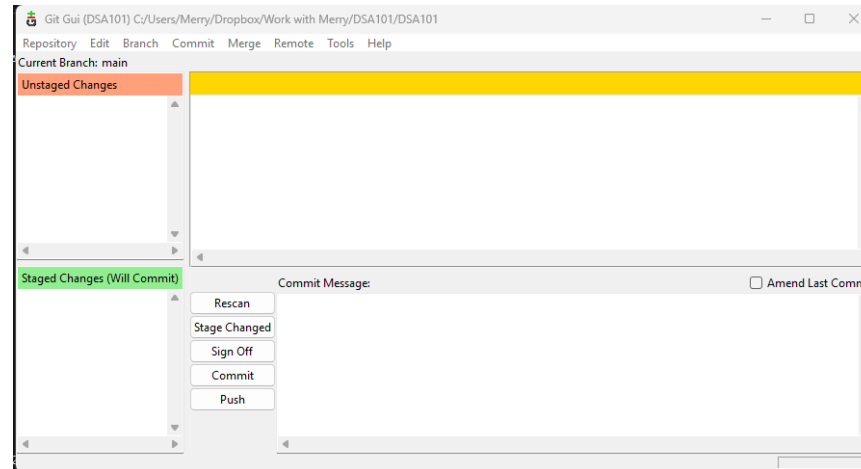
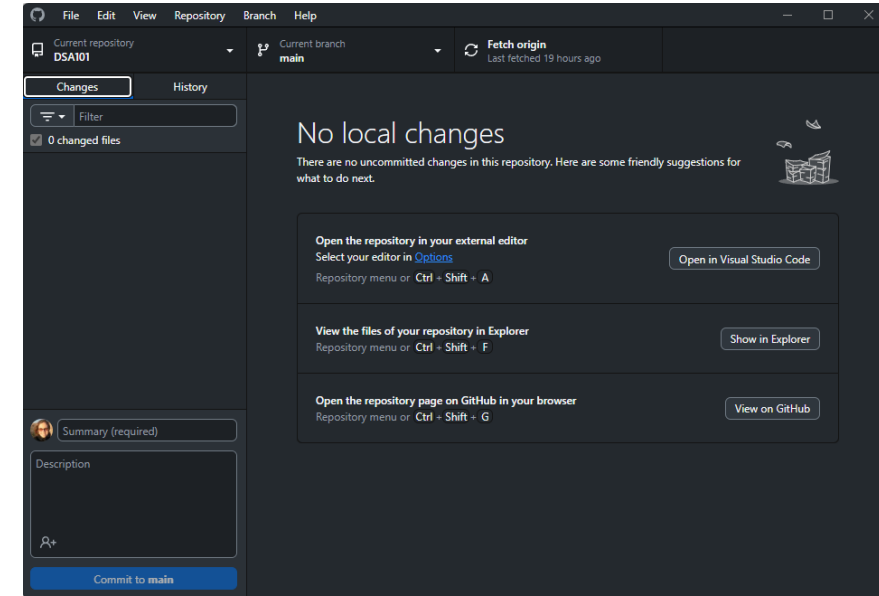
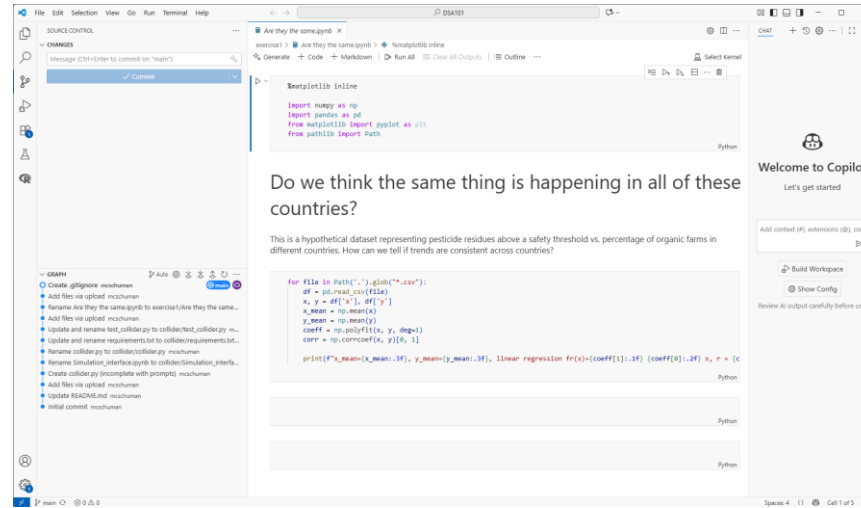
For compliance reasons, we're required to collect country information to send you occasional updates and announcements.

☐ Receive occasional product updates and announcements

By creating an account, you agree to the [Terms of Service](#). For more information about GitHub's privacy practices, see the [GitHub Privacy Statement](#). We'll occasionally send you account-related emails.

Let's start!

- (1) [Make a GitHub account](#) if you don't have one
- (2) [Install Git](#): there are many options!



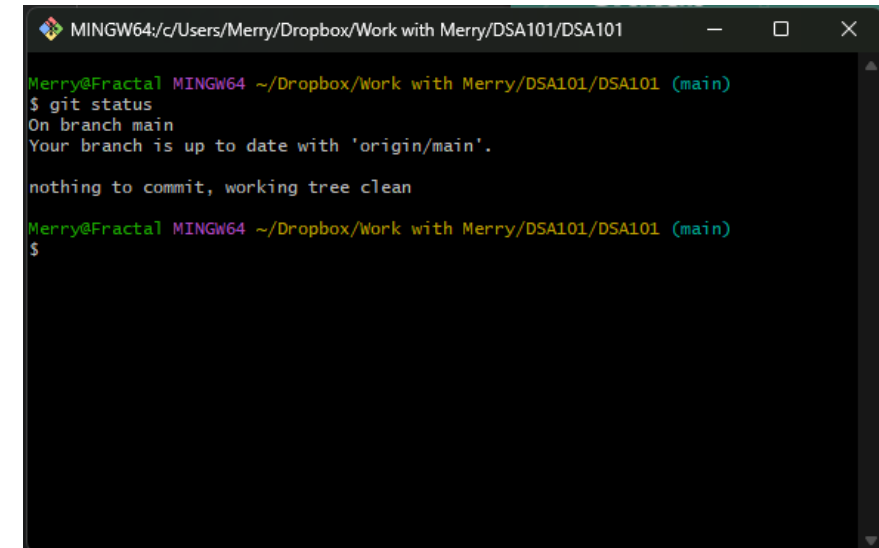
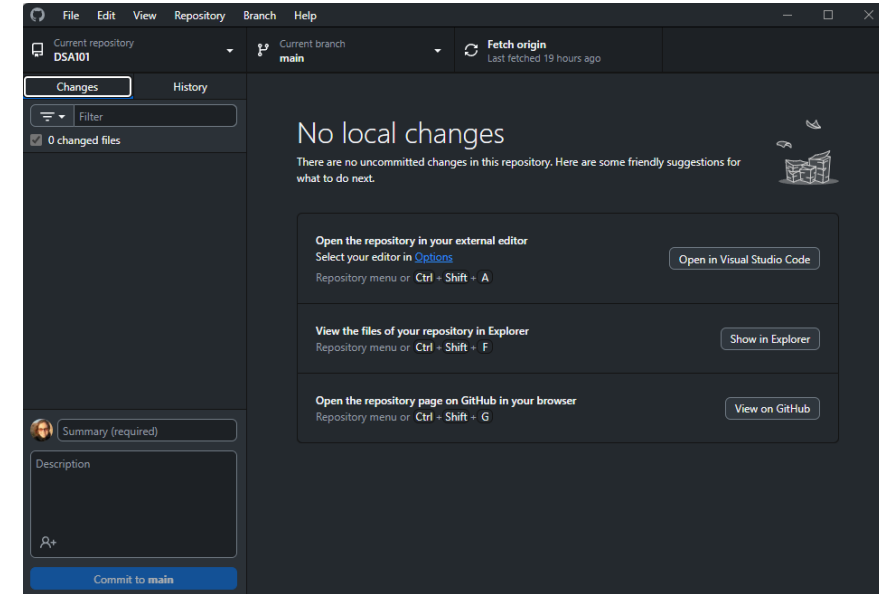
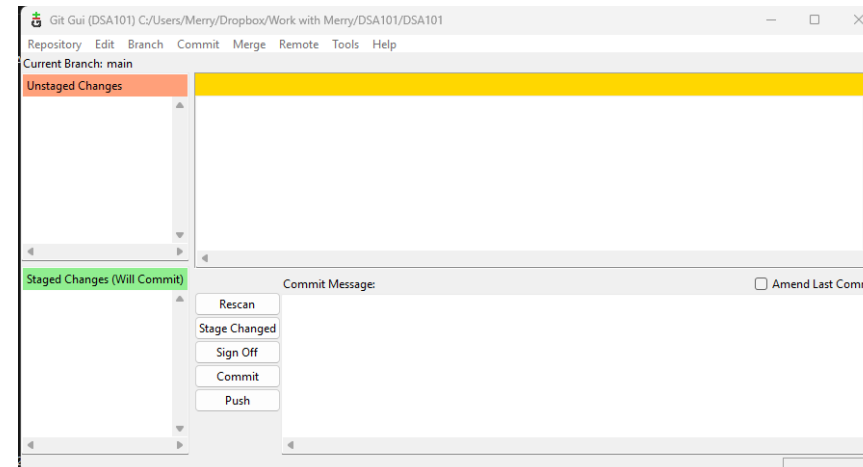
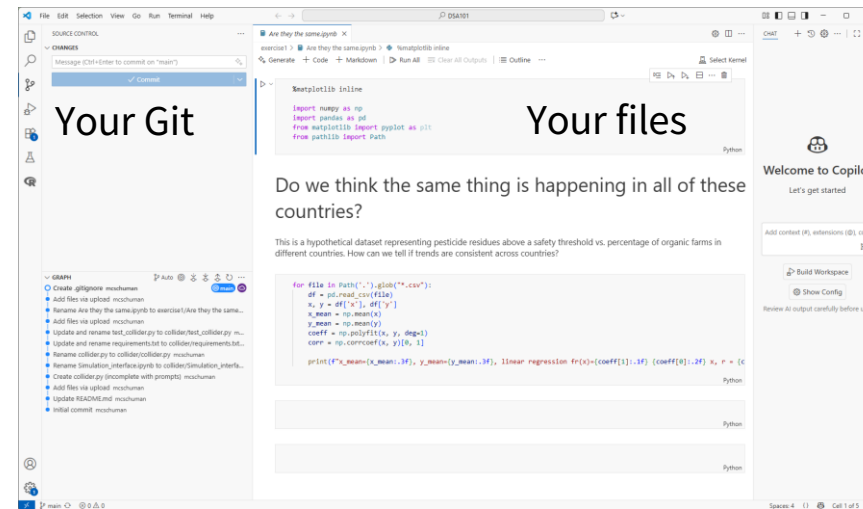
Let's start!

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many options!

- **Easiest:** an IDE like Visual Studio Code: interface to interact with code and Git

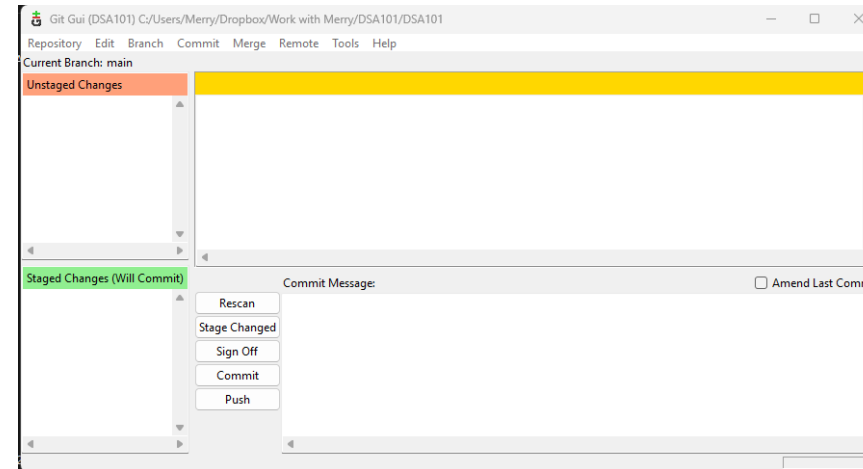
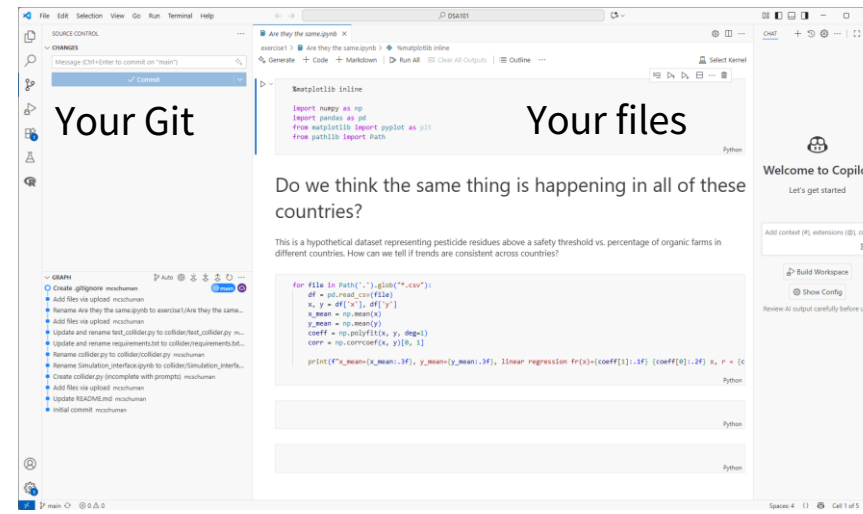
Visual Studio Code or another IDE



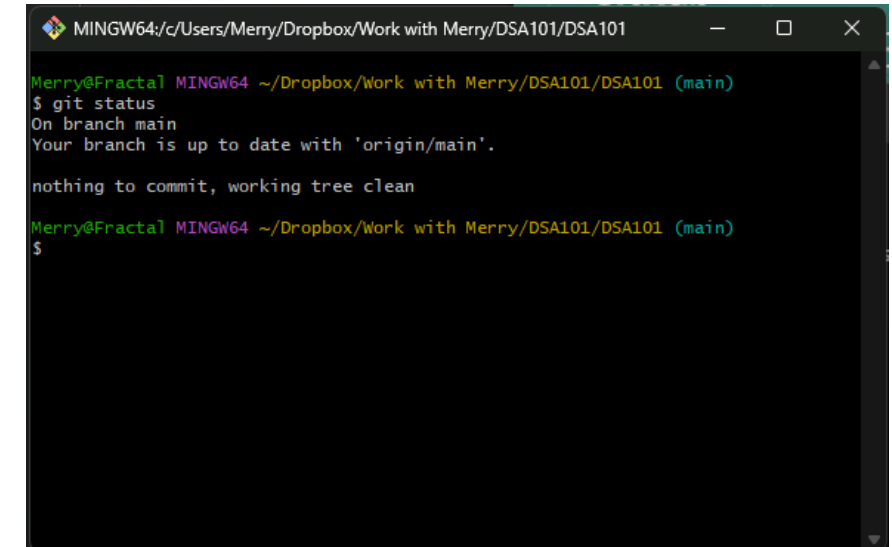
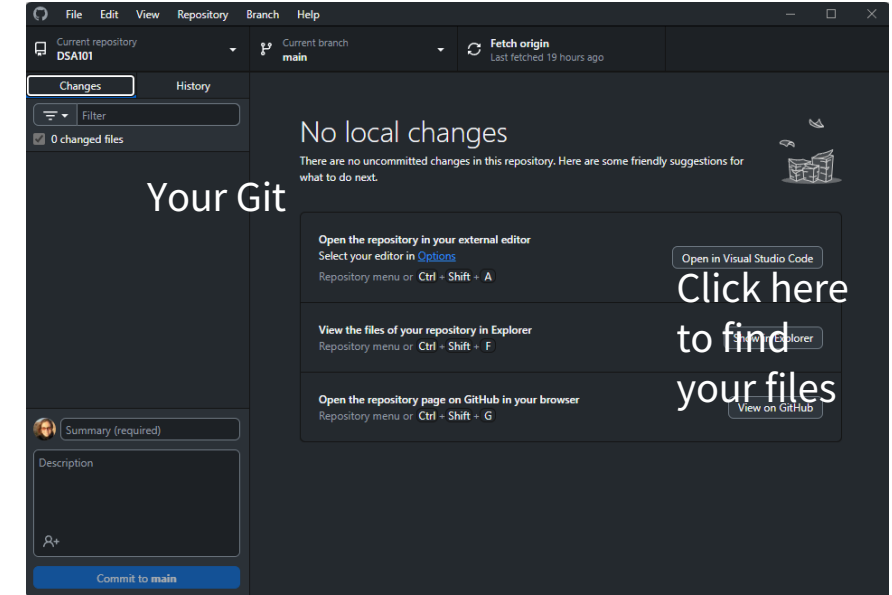
Let's start!

- (1) [Make a GitHub account](#) if you don't have one
- (2) [Install Git](#): there are many options!
 - **Easiest**: an IDE like Visual Studio Code: interface to interact with code and Git
 - **Second easiest**: GitHub Desktop, a GUI that prompts you to use Git correctly

Visual Studio Code or another IDE



GitHub Desktop



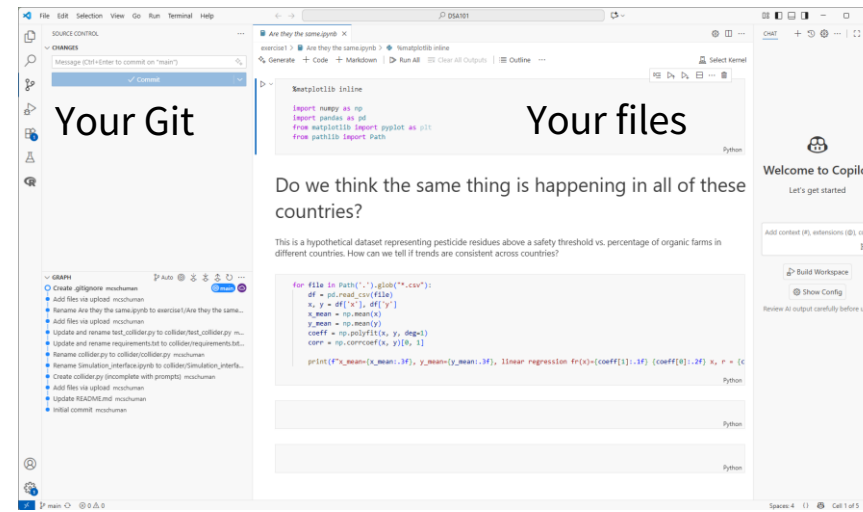
Let's start!

(1) [Make a GitHub account](#) if you don't have one

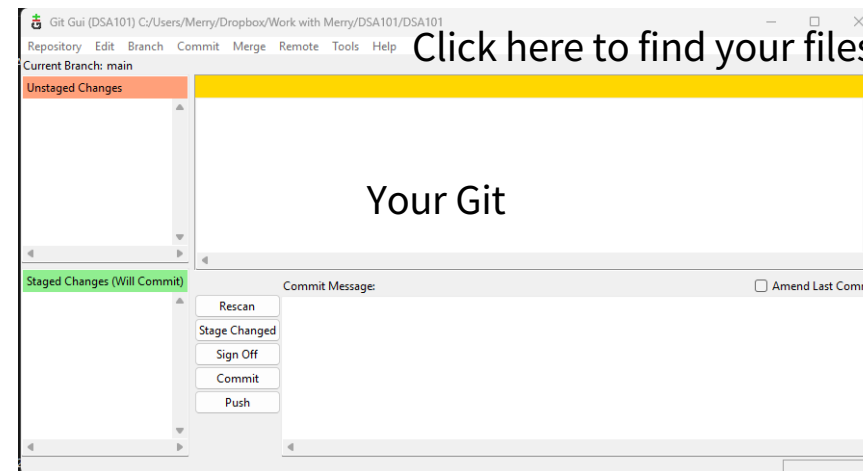
(2) [Install Git](#): there are many options!

- **Easiest:** an IDE like Visual Studio Code: interface to interact with code and Git
- **Second easiest:** GitHub Desktop, a GUI that prompts you to use Git correctly
- **Develop your computer skills (still fairly easy):** install with Git installer, then interact with Git using Git GUI or Git Bash (or Terminal)

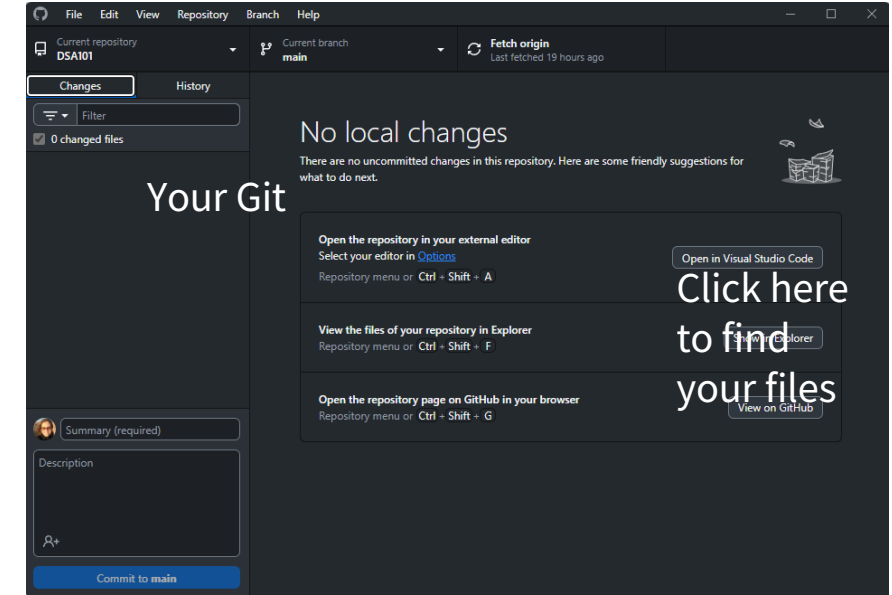
Visual Studio Code or another IDE



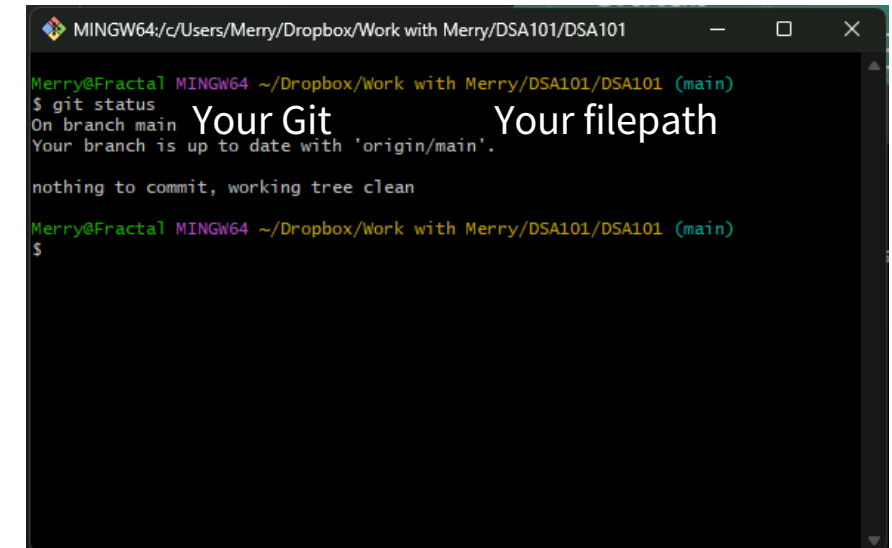
Git GUI



GitHub Desktop

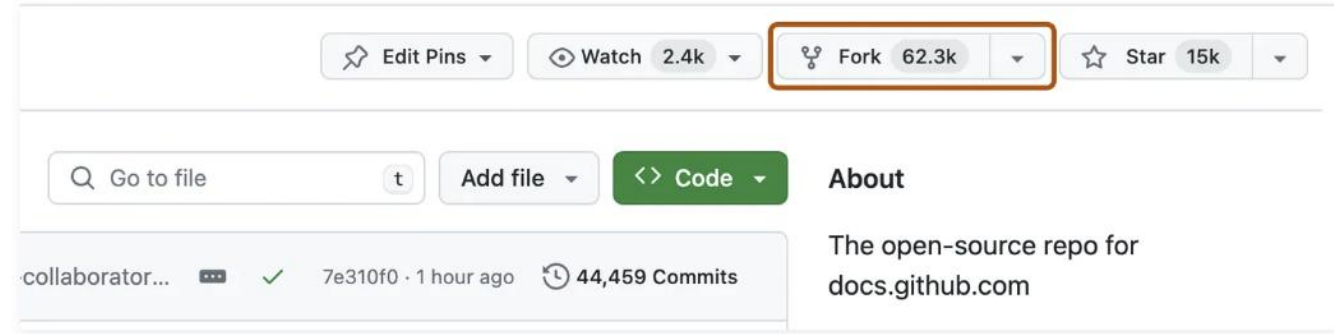


Git Bash



Let's start!

- (1) [Make a GitHub account](#) if you don't have one
- (2) [Install Git](#): there are many options!
- (3) [Fork](#) the DSA101 repository



- 3 Under "Owner," select the dropdown menu and click an owner for the forked repository.
- 4 By default, forks are named the same as their upstream repositories. Optionally, to further distinguish your fork, in the "Repository name" field, type a name.
- 5 Optionally, in the "Description" field, type a description of your fork.
- 6 Optionally, select **Copy the DEFAULT branch only**.

For many forking scenarios, such as contributing to open-source projects, you only need to copy the default branch. If you do not select this option, all branches will be copied into the new fork.

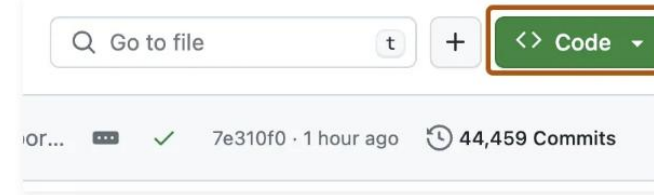
- 7 Click **Create fork**.

Let's start!

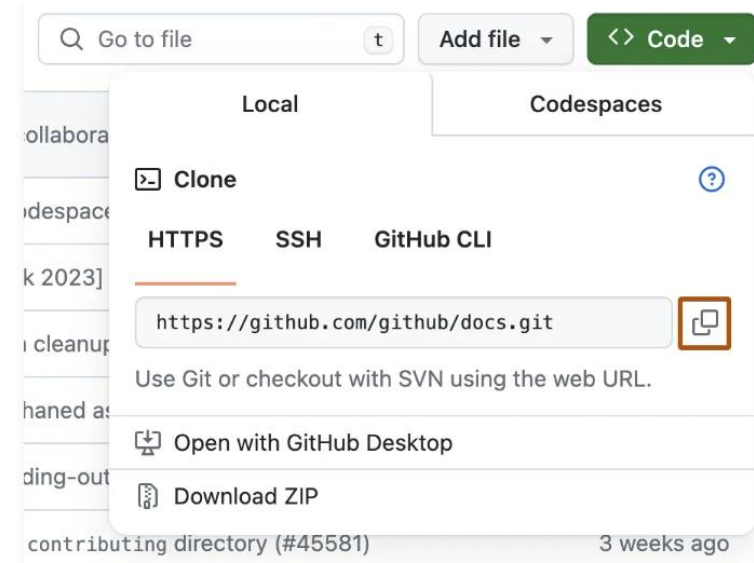
- (1) [Make a GitHub account](#)
if you don't have one
- (2) [Install Git](#): there are many options!
- (3) [Fork](#) the DSA101 repository
- (4) [Clone your fork](#)

I expect you to understand how Git works at this level for the exam.

<https://docs.github.com/en/pull-requests/collaborating-with-pull-requests/working-with-forks/fork-a-repo>



- 3 Copy the URL for the repository.
 - To clone the repository using HTTPS, under "HTTPS", click .
 - To clone the repository using an SSH key, including a certificate issued by your organization's SSH certificate authority, click **SSH**, then click .
 - To clone a repository using GitHub CLI, click **GitHub CLI**, then click .



Next: Try exercise1

Are they the same?

We have a fictitious dataset from 12 different countries documenting the relationship between proportion of organic farms, and proportion of test sites in which pesticide residues are above a regulatory threshold.

You will characterize this relationship for each country and answer the question, is the relationship essentially the same across all countries in our dataset?

I expect you to understand how to look at data to answer specific questions for the exam.



<https://4.bp.blogspot.com/-1gcbXO2ePh4/UY8Vzp5khqI/AAAAAAAAJbk/i6HUCVq43GE/s1600/pesticide+spraying+man.jpg>